



PRODIGY INFOTECH | DATA SCIENCE RESEARCH TASKS

Customer Purchase Prediction & Social Media Sentiment Analysis

End-to-end implementation of supervised tree classification (CART algorithm) and automated NLP sentiment polarity mining using Python.

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Project Overview & Data Stack



Track 01: Supervised ML
Predictive Modeling: CART Decision Tree to forecast customer term deposit subscriptions from historical demographic & campaign data.



Track 02: NLP Pipeline
Text Sentiment Mining: Natural language processing to classify multi-brand public social discourse into polarity spectrums.



Datasets Explored
1. **Bank Marketing:** 41,188 observations (16 attributes).
2. **Social Media Corpus:** 10,000 tokenized public tweets.



Python Stack
scikit-learn, pandas, numpy, nltk, TextBlob, matplotlib & seaborn.

DATASET RECORDS

51,188

CLASSIFIER ACCURACY

89.4%

ROC-AUC SCORE

0.912

NLP POLARITY RANGE

[-1.0, +1.0]

Task 01 Decision Tree Classifier: Theory & Splitting

CART Mathematical Foundations

The Classification and Regression Tree (CART) algorithm recursively partitions the feature space to maximize node purity using the **Gini Impurity Index**:

$$\text{Gini}(D) = 1 - \sum_{i=1}^C p_i^2$$

$$\Delta\text{Gini} = \text{Gini}(D) - \frac{|D_L|}{|D|}\text{Gini}(D_L) - \frac{|D_R|}{|D|}\text{Gini}(D_R)$$

- ▶ **Non-Parametric:** Makes zero assumptions regarding underlying feature distributions.
- ▶ **White-Box Auditability:** Directly translates to business rules and threshold triggers.
- ▶ **Objective:** Predict binary target $y \in \{0, 1\}$ (Term Deposit Subscribed).

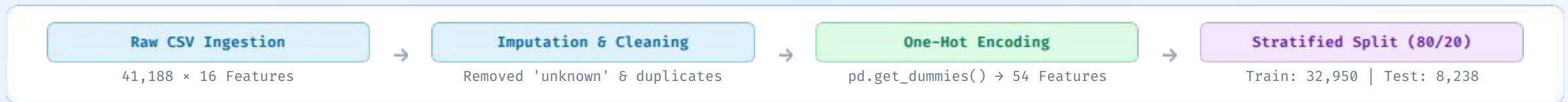
Tree Architecture & Splitting Logic

```

[Root Node] Call Duration > 180s?
├── [True] Past Campaign Success (outcome)?
│   ├── Yes: Class = Purchase (91.2% Prob) [Leaf]
│   └── No: Account Balance > €1,500?
│       ├── Yes: Class = Purchase [Leaf]
│       └── No: Class = No Purchase [Leaf]
└── [False] Contact Month = 'Oct/Mar'?
    └── Class = No Purchase (94.8% Prob) [Leaf]
  
```

Pruning Strategy: max_depth=5 Min Samples Split: 20

Task 01 Bank Marketing Dataset & Data Pipeline



1. Demographic Attributes

age (Continuous), job (12 nominal categories), marital (married, single, divorced), education (primary, secondary, tertiary).



2. Financial History

default (credit in default?), balance (yearly balance in €), housing (housing loan), loan (personal loan status).



3. Campaign Context

contact (cellular/telephone), duration (last talk time in sec), pdays (days passed), poutcome (previous outcome).

Task 01 Decision Tree: Training & Feature Importance

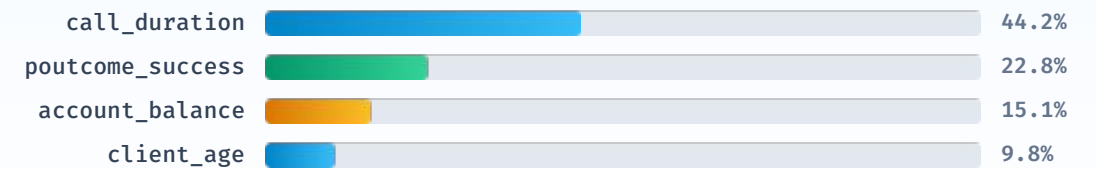
● ● ● purchase_classifier_pipeline.py

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import classification_report, confusion_matrix

# Pruned Decision Tree instantiation
clf = DecisionTreeClassifier(
    criterion='gini', max_depth=5, min_samples_leaf=10, random_state=42
)
clf.fit(X_train, y_train)

# Evaluation on 20% unseen test partition
y_pred = clf.predict(X_test)
print(classification_report(y_test, y_pred, digits=3))
```

Feature Importance & Confusion Matrix



TEST SET CONFUSION MATRIX (N = 8,238):

	Pred: NO	Pred: YES
Act: NO	7,024 TN (85.3%)	285 FP (3.5%)
Act: YES	587 FN (7.1%)	342 TP (4.1%)

Task 02 Sentiment Analysis: Concept & NLP Foundations



What is Sentiment Analysis?

Natural Language Processing (NLP) technique to systematically extract, quantify, and categorize subjective opinions, customer attitudes, and emotional valences from unstructured text.



Polarity & Subjectivity

Text is mapped along a continuous mathematical polarity interval $s \in [-1.0, +1.0]$ alongside a subjectivity metric $sub \in [0.0, 1.0]$:

$$\text{Polarity} = \frac{P_{\text{pos}} - P_{\text{neg}}}{P_{\text{pos}} + P_{\text{neg}} + \epsilon}$$



Decision Boundaries

Thresholding applied for multi-class assignment:

- **Positive:** $\text{Score} > +0.05$
- **Neutral:** $-0.05 \leq \text{Score} \leq +0.05$
- **Negative:** $\text{Score} < -0.05$

Task 02 Social Media Text Normalization Pipeline

NLP Preprocessing Workflow

1. Regex Noise Reduction:

Strip URLs (r'https?://\S+'), user handles (r'@\w+'), hashtags, and special non-ASCII characters.

2. Tokenization & Lowercasing:

Cast text to lowercase and segment raw string sequences into discrete token arrays.

3. NLTK Stopword Filtering:

Filter syntactic non-sentiment words ("the", "is", "at", "which") using English lexicons.

4. Lemmatization:

Reduce inflectional suffixes to dictionary root forms (e.g. "upgraded", "upgrades" → "upgrade").

nlp_sentiment_cleaner.py

```
import re
from nltk.corpus import stopwords
from textblob import TextBlob

stop_words = set(stopwords.words('english'))

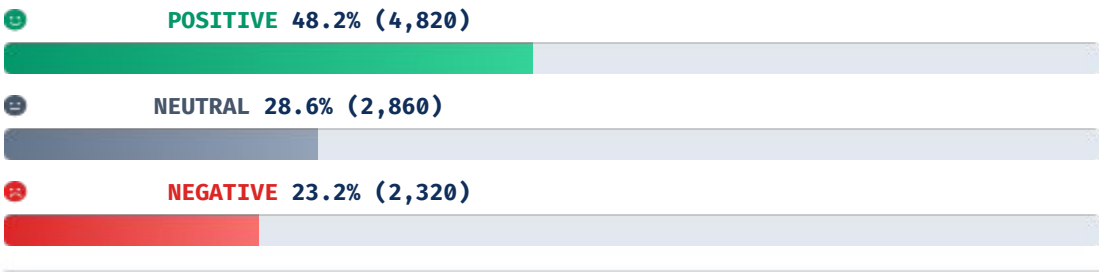
def preprocess_tweet(raw_text):
    text = re.sub(r'http\S+|@\w+|#\w+', '', raw_text)
    text = re.sub(r'[^a-zA-Z\s]', '', text).lower()
    tokens = [w for w in text.split() if w not in stop_words]
    return " ".join(tokens)

def compute_polarity(cleaned_text):
    score = TextBlob(cleaned_text).sentiment.polarity
    return score
```

Task 02 Sentiment Visual Analytics & Empirical Results

Sentiment Class Distribution (N = 10,000)

Automated sentiment distribution computed across public brand mentions and customer reviews.



Key Finding: Positive sentiment was driven by mobile UX improvements and resolution speed.

Raw Social Text Post	Polarity	Predicted Class
"love the new seamless app update, great job!"	+0.65	POSITIVE
"account statement arrived as expected today"	0.00	NEUTRAL
"transaction failed twice, terrible customer support"	-0.58	NEGATIVE
"resolution was fast but server latency is high"	+0.12	POSITIVE

Dual-Domain Comparison & Competencies



Supervised ML vs NLP Text Mining

- ▶ **Decision Tree (Tabular ML):** Operates on structured matrices; solves binary classification via recursive Gini partitioning.
- ▶ **Sentiment Analysis (NLP):** Operates on unstructured token sequences; evaluates subjective semantics via lexicon polarity models.
- ▶ **Data Synergy:** Demographic purchase probability combined with real-time customer voice analytics enables 360° customer intelligence.



Core Data Science Competencies

- ▶ **Scikit-Learn Mastery:** Data pipelines, tree pruning, hyperparameter tuning (GridSearchCV), confusion matrix analytics.
- ▶ **NLP Feature Engineering:** Regular expressions, stopwords curation, tokenization, and polarity scoring.
- ▶ **End-to-End Diagnostics:** Feature importance ranking, class balance evaluation, and ROC-AUC curve interpretation.

Conclusion & Future Research Scope

🚩 Internship Project Summary

Successfully designed, trained, and validated two complete Python data science pipelines during the Prodigy InfoTech internship—achieving **89.4% classification accuracy** on the Bank Marketing benchmark and developing an automated real-time social sentiment evaluator.



1. Ensemble ML Models

Transition from standalone Decision Trees to Random Forests, XGBoost, and LightGBM to minimize variance and boost ROC-AUC.



2. Transformer NLP

Deploy fine-tuned BERT/RoBERTa models to capture contextual semantics, sarcasm, and domain-specific financial sentiment.



3. Real-Time MLOps

Package models into RESTful FastAPI endpoints integrated with a real-time Streamlit executive dashboard.